

Palle Jacob Abraham Jane

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LinkedIn: www.linkedin.com/in/pjacobjane | Website: https://jacobabrahamjane.github.io/JACOB_Portfolio/

GitHub: <https://github.com/JACOBABRAHAMJANE/PROJECTS.git>

Technical Skills

Languages & Cloud: Python, SQL, Java, C, Google Cloud Platform (GCP).

Data Analytics & ML: Tableau, Excel, Pandas, NumPy, Statistics, EDA, Random Forest, XGBoost, LightGBM, CatBoost.

Tools & Frameworks: Git, GitHub, Figma, Android SDK.

DSA Questions: Solved 150+ DSA questions at LeetCode.

Education

R.V.R. & J.C. College of Engineering, Guntur **2022 - 2026**

Bachelor of Technology in Computer Science & Engineering (Data Science)

CGPA: 7.36

Scaler Academy – Advanced Software & Data Science Acceleration Program **2025 - 2026**

(Advanced Data Structures, Algorithms, Machine Learning & Analytics Engineering)

One among Top 3 in the leader board

Core projects at R.V.R. & J. C.

IoT Device Security Framework | *LightGBM, CatBoost, Random Forest*

- Designed a hybrid-ensemble intrusion detection system utilizing LightGBM, CatBoost, and Random Forest, systematically evaluated across a benchmark dataset of **1,75,341 network traffic records**.
- Optimized classification pathways by processing **49 high-dimensional features** to classify **10 distinct threat vectors** (including DoS and Botnets), maintaining a stable **97.9% CRNN baseline accuracy** during model cross-validation.

Employee Attrition Prediction System | *XGBoost, LightGBM, SHAP*

- Engineered a hybrid machine learning pipeline using an **80:20 stratified split** on structured HR data, deploying SMOTE oversampling to mitigate heavy class imbalances.
- Implemented a soft-voting ensemble combining LightGBM and CatBoost with a Genetic Algorithm for feature selection, driving a **6% improvement in F1-Score (reaching 93%)** and achieving a **98% ROC-AUC evaluation metric**.

Analytics Case Studies (SCALER)

- Target (SQL):** Engineered complex queries across **100,000+ e-commerce transaction rows** to isolate seasonal purchasing spikes and compute month-on-month revenue growth across Brazilian states.
- Delhivery (Python):** Processed a telemetry logistics dataset of **17,786 rows**, engineering spatial-temporal features and executing dual-sample t-tests (**t=26.69, t=10.58**) to isolate routing algorithm variances.
- Walmart (Python):** Analyzed a Black Friday dataset of **550,068 rows**, implementing Central Limit Theorem workflows and **95% Confidence Intervals** to mathematically validate non-overlapping gender purchasing variances.
- Yulu (Python):** Formulated ANOVA (**F=236.95**) and Chi-Square (**chi2=49.16**) testing on **10,886 rental records** to map vehicle demand shifts across seasonal and weather matrices.
- Netflix (Python):** Evaluated content metadata distribution patterns across **8,807 titles**, unnesting comma-separated categorical text variables to profile categorical production dominance across **10 unique global zones**.

Internships

GenAI & Cloud Analytics Intern | *Google Cloud Platform (GCP) | Remote*

- Deployed enterprise-scale LLMs for text summarization while optimizing prompt infrastructure.
- Enhanced inference workloads, reducing cloud computational overhead while maintaining high precision.

Android Development Intern | *Google Virtual Internship | Remote*

- Implemented core Android SDK frameworks to design highly responsive interface architectures.
- Debugged and refactored code modules to systematically improve runtime execution performance.

Certifications

- Certificate on **"100 Days of Code: Python Pro Bootcamp"** from Udemy.
- Certificate on **"Programming in Java"** from NPTEL, IIT Madras.
- Certificate on **"Quantum Fundamentals & Advanced Algorithms"** from Qubitech.